MSG No. 60.431 P_Cbcn [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,5,10,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,6,9,13]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,8,11,15]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,5,10,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,6,9,13]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,8,11,15]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3,14,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[2,4,13,15]`
3	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[6,8,9,11]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[1,13]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[2,14]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,15]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[4,16]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[5,9]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[6,10]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,11]`
8	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[8,12]`

Table 5: Wyckoff site: `8e`, site symmetry: `2'..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,9]`
3	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,11]`

continued ...

Table 5

No.	position	mapping
5	$[-x,\ 0,\ 0]$	`[5,14]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[6,13]`
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[8,15]`

Table 6: Wyckoff site: `8f`, site symmetry: `m'..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[2,13]`
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	`[3,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[4,15]`
5	$[0,\ -y,\ -z]$	`[5,10]`
6	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[6,9]`
7	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[7,12]`
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[8,11]`

Table 7: Wyckoff site: `8g`, site symmetry: `..m'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,16]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[2,15]`
3	$\left[-x,\ y,\ \frac{1}{4}\right]$	`[3,14]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[4,13]`
5	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[5,12]`
6	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[6,11]`
7	$\left[x,\ -y,\ \frac{3}{4}\right]$	`[7,10]`
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[8,9]`

Table 8: Wyckoff site: `16h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[2]`
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[6]`
7	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9]`
10	$[x,\ -y,\ -z]$	`[10]`
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11]`

continued ...

Table 8

No.	position	mapping
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[13]
14	$[-x,\, y,\, z]$	[14]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[16]